HAZE ON THE HORIZON:DELHI VS. CHANDIGARH

Multi-Factor Analysis Of Air Quality (Delhi Vs. Chandigarh)

The Objective

To quantify the impact of Aviation✈️, Seasons❄️,Festivals on the India's capital and a controlled city(Chandigarh)

Key Findings (The 4 Factors)

1.Aviation

Peak AQI(Nov-Jan) aligns directly with IGI Airport as it contributes in No2 the most.

2.Traffic

No significant dip on weekends proves heavy industry/trucks >commute cars

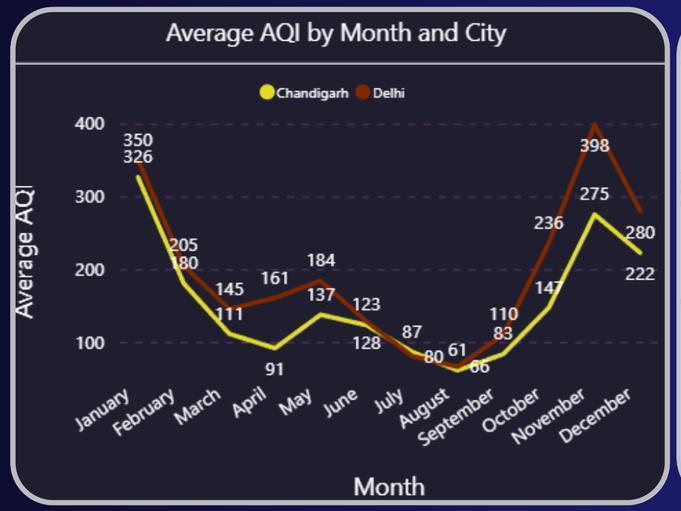
3.Seasons

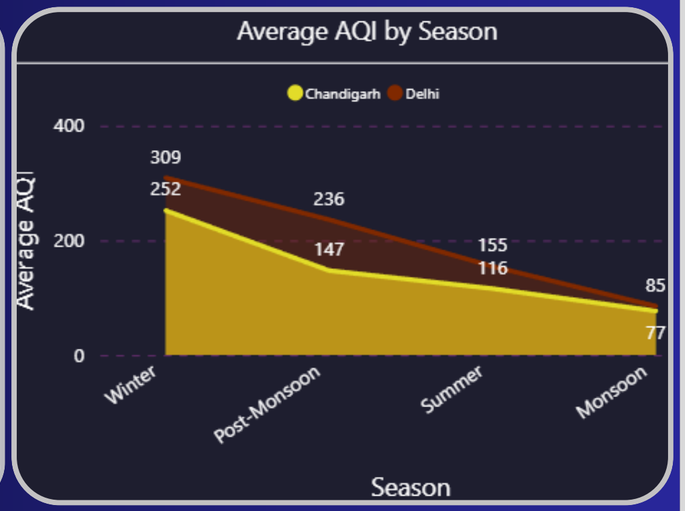
Winters acts as a "pollution multiplier" for both the cities but Delhi suffers 30% higher intensity

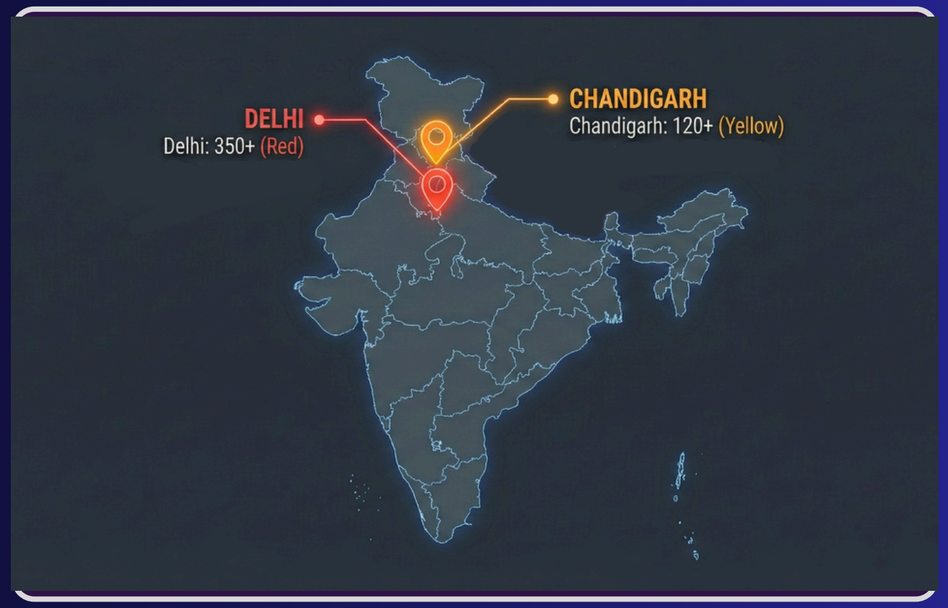
4.Festivals

Festivals creates massive but short-lived 72-hour hazzard spike.

Visual Analysis







Abstract

Abstract This study analyzes daily AQI variations in Delhi and Chandigarh (2024) using CPCB data to isolate the impact of urbanization on air quality. By evaluating seasonal trends, festival spikes, and traffic patterns, the research reveals that while both cities share regional weather struggles, Delhi's industrial density results in a 27% higher average pollution load. Findings indicate that winter inversion layers and continuous base-load emissions (rather than just weekend commuter traffic) are the primary drivers of the capital's hazardous air quality.

Conclusion

Conclusion Analysis identifies Winter as the critical driver, spiking Delhi's average AQI to 310 (a 27% increase over Chandigarh). Crucially, elevated NO_2 concentrations point to the aviation and transport sectors as major contributors to the dense smog that hampers visibility. This accumulation of gases confirms that while seasonal inversion acts as the trigger, the persistent base load from these sectors is the root cause of the crisis.